

EAS 501 Lecture Notes 09/02/2026

Cheng Wang

Mathematics @ UMass Dartmouth

Outline

- Chapter 13. Differential Calculus and Functions of Several Variables

13.2 Preliminaries

For $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$, $\mathbf{x}' = (x'_1, x'_2, \dots, x'_n)^T$, the Euclidean norm of $\mathbf{x}' - \mathbf{x}$ is defined as

$$|\mathbf{x}' - \mathbf{x}| := \sqrt{(x'_1 - x_1)^2 + (x'_2 - x_2)^2 + \dots + (x'_n - x_n)^2}.$$

Triangle inequality: $|\vec{\mathbf{a}} + \vec{\mathbf{b}}| \leq |\vec{\mathbf{a}}| + |\vec{\mathbf{b}}|$

Continuity: $f(\mathbf{x})$ is continuous at $\mathbf{x} = \mathbf{x}_0$ if

$\lim_{\mathbf{x} \rightarrow \mathbf{x}_0} f(\mathbf{x}) = f(\mathbf{x}_0)$, i.e., for each $\varepsilon > 0$, there corresponds a $\delta(\varepsilon, \mathbf{x}_0) > 0$ such that, $|f(\mathbf{x}) - f(\mathbf{x}_0)| < \varepsilon$ whenever $|\mathbf{x} - \mathbf{x}_0| < \delta$.

13.3 Partial derivatives

Consider a two-dimensional (2-D) function $f(x, y)$, and a fixed point (x_0, y_0)

$$f_x(x_0, y_0) = \frac{\partial f}{\partial x}(x_0, y_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x}, \quad (1)$$

$$f_y(x_0, y_0) = \frac{\partial f}{\partial y}(x_0, y_0) = \lim_{\Delta y \rightarrow 0} \frac{f(x_0, y_0 + \Delta y) - f(x_0, y_0)}{\Delta y}. \quad (2)$$

In addition, we define $f_{xx} = \frac{\partial}{\partial x}(f_x)$, $f_{xy} = \frac{\partial}{\partial y}(f_x)$, $f_{yx} = \frac{\partial}{\partial x}(f_y)$, $f_{yy} = \frac{\partial}{\partial y}(f_y)$, $f_{yyx} = \frac{\partial}{\partial x}(f_{yy})$, $f_{xyy} = \frac{\partial}{\partial y}(f_{xy})$, etc.

Example 1. $f(x, y) = \begin{cases} \frac{xy(x^2 - y^2)}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0), \\ 0, & \text{if } (x, y) = (0, 0), \end{cases}$

A careful calculation implies that

$$f_x = \begin{cases} \frac{y(x^4 + 4x^2y^2 - y^4)}{(x^2 + y^2)^2}, & \text{if } (x, y) \neq (0, 0), \\ 0, & \text{if } (x, y) = (0, 0), \end{cases}$$

$$f_y = \begin{cases} \frac{x(x^4 - 4x^2y^2 - y^4)}{(x^2 + y^2)^2}, & \text{if } (x, y) \neq (0, 0), \\ 0, & \text{if } (x, y) = (0, 0), \end{cases}$$

In turn, we get

$$\begin{aligned} f_{xy}(0, 0) &= \lim_{\Delta y \rightarrow 0} \frac{f_x(0, \Delta y) - f_x(0, 0)}{\Delta y} \\ &= \lim_{\Delta y \rightarrow 0} \frac{-\Delta y - 0}{\Delta y} = -1, \end{aligned}$$

$$f_{yx}(0, 0) = \lim_{\Delta x \rightarrow 0} \frac{f_y(\Delta x, 0) - f_y(0, 0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\Delta x - 0}{\Delta x} = 1.$$

Theorem 13.3.1 Equality of mixed partial derivatives

If f_x , f_y , f_{xy} and f_{yx} are continuous in some neighborhood of (x_0, y_0) , then $f_{xy}(x_0, y_0) = f_{yx}(x_0, y_0)$.

Example 13.3 # 1 (e) $f(x, y) = (x^2 + y^2)^{\frac{1}{2}}$

A careful calculation reveals that

$$f_x = \begin{cases} \frac{x}{(x^2+y^2)^{\frac{1}{2}}}, & \text{if } (x, y) \neq (0, 0), \\ 0, & \text{if } (x, y) = (0, 0), \end{cases}$$

$$f_y = \begin{cases} \frac{y}{(x^2+y^2)^{\frac{1}{2}}}, & \text{if } (x, y) \neq (0, 0), \\ 0, & \text{if } (x, y) = (0, 0), \end{cases}$$

Moreover, we have $f_{xy} = f_{yx} = -\frac{xy}{(x^2+y^2)^{\frac{3}{2}}}$, if $(x, y) \neq (0, 0)$.

It is clear that neither f_x , f_y , nor f_{xy} , f_{yx} is continuous at $(0, 0)$, while $f_{xy}(0, 0) = f_{yx}(0, 0) = 0$.

13.4 Composite functions and chain differentiations

For one-variable function: If $F(t) = f(x(t))$, we have

$$\frac{dF}{dt} = \frac{df}{dx} \cdot \frac{dx}{dt}.$$

For multi-variable functions: If $F(t) = f(x_1(t), x_2(t), \dots, x_n(t))$, we have

$$\frac{dF}{dt} = \frac{\partial f}{\partial x_1} \cdot \frac{dx_1}{dt} + \frac{\partial f}{\partial x_2} \cdot \frac{dx_2}{dt} + \dots + \frac{\partial f}{\partial x_n} \cdot \frac{dx_n}{dt}.$$

13.5 Taylor's formula and mean value theorem

For a single-variable function $f(x)$, the Taylor series are given by

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \dots \\ + \frac{f^{(n-1)}(a)}{(n-1)!}(x - a)^{n-1} + R_n(x), \quad (3)$$

$$\textbf{Lagrange form : } R_n(x) = \frac{f^{(n)}(\xi)}{n!}(x - a)^n, \quad (4)$$

$$\textbf{Integral form : } R_n(x) = \int_a^x \frac{f^{(n)}(t)}{(n-1)!}(x - t)^{n-1} dt, \quad (5)$$

in which ξ is between a and x .

Example 13.5 # 4 (b)

Given $f(x) = \sin x$, its Taylor series are given by

$$\sin x = x - \frac{x^3}{3!} + \frac{\cos \xi}{5!}x^5, \quad \xi \text{ between } 0 \text{ and } x.$$

For $0 \leq x \leq 0.5$, what is the maximum error of $|\sin x - P_3(x)|$?

Solution. We have (by noticing that $|\cos \xi| \leq 1$)

$$\begin{aligned} |\sin x - P_3(x)| &= \left| \sin x - \left(x - \frac{x^3}{3!} \right) \right| = \left| \frac{\cos \xi}{5!}x^5 \right| \\ &= \frac{|\cos \xi|}{120} \cdot |x|^5 \leq \frac{1}{120} \cdot 0.5^5 \approx 0.000261. \quad (6) \end{aligned}$$

$$\sin x = x - \frac{x^3}{3!} + \dots + \frac{(-1)^{k-1}}{(2k-1)!}x^{2k-1} + \frac{(-1)^k \cos \xi}{(2k+1)!}x^{2k+1}, \quad (7)$$

$$\cos x = 1 - \frac{x^2}{2!} + \dots + \frac{(-1)^k}{(2k)!}x^{2k} + \frac{(-1)^{k+1} \cos \xi}{(2k+2)!}x^{2k+2}. \quad (8)$$

Theorem 13.5.1 Mean value theorem

Let f be continuous on $[a, x]$ and differentiable on (a, x) , then there exists $\xi \in (a, x)$ so that $f(x) = f(a) + f'(\xi)(x - a)$.

2-D version: Let $f(x, y)$ and f_x, f_y be continuous in an open region R so that $(a, b), (x, y) \in R$, and the line from (a, b) to (x, y) lies inside R . Then there exists (ξ, η) on that line, with $f(x, y) = f(a, b) + f_x(\xi, \eta)(x - a) + f_y(\xi, \eta)(y - b)$.

In this case, the 2-D Taylor series are given by

$$\begin{aligned} TS f|_{(a,b)} &= f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b) \\ &\quad + \frac{1}{2!} \left(f_{xx}(a, b)(x - a)^2 \right. \\ &\quad \left. + 2f_{xy}(a, b)(x - a)(y - b) \right. \\ &\quad \left. + f_{yy}(a, b)(y - b)^2 \right) + \dots \end{aligned}$$

13.6 Implicit functions and Jacobians

Theorem 13.6.1 **Implicit function theorem, single variable case** Let $f(x_0, y_0) = 0$, and $f(x, y)$ is C^1 in some neighborhood of (x_0, y_0) with $f_y(x_0, y_0) \neq 0$. Then $f(x, y) = 0$ uniquely implies a function $y(x)$ in some neighborhood of x_0 , with $y(x_0) = y_0$, and $y(x)$ is differentiable.

Theorem 13.6.2 Implicit function theorem, multi variable case

$$(*) \quad \text{For } \begin{cases} f_1(x_1, \dots, x_n, u_1, \dots, u_n) = 0, \\ \dots \\ f_n(x_1, \dots, x_n, u_1, \dots, u_n) = 0, \end{cases}$$

satisfied by $(x_{1,0}, \dots, x_{n,0}, u_{1,0}, \dots, u_{n,0})$, and $f_1, f_2, \dots, f_n$ are C^1 with $\det J|_{(x_{1,0}, \dots, x_{n,0}, u_{1,0}, \dots, u_{n,0})} \neq 0$,

$$J = \begin{pmatrix} \frac{\partial f_1}{\partial u_1} & \frac{\partial f_1}{\partial u_2} & \dots & \frac{\partial f_1}{\partial u_n} \\ \dots & \dots & \dots & \dots \\ \frac{\partial f_n}{\partial u_1} & \frac{\partial f_n}{\partial u_2} & \dots & \frac{\partial f_n}{\partial u_n} \end{pmatrix}.$$

Then $(*)$ uniquely implies functions $u_1(x_1, \dots, x_n), \dots, u_n(x_1, \dots, x_n)$ in some neighborhood of $(x_{1,0}, \dots, x_{n,0}, u_{1,0})$, and the function is C^1 .

Example 13.6 # 4 (b)

$$\begin{cases} x - u \cos v = 0, \\ y - u \sin v = 0, \end{cases} \quad \text{given } (x_0, y_0, u_0, v_0) = (0, 2, 2, \frac{\pi}{2}).$$

Do implicit functions $u(x, y)$ and $v(x, y)$ exist in some neighborhood of $(x_0, y_0) = (0, 2)$?

Solution. We denote $f_1(x, y, u, v) = x - u \cos v$, $f_2(x, y, u, v) = y - u \sin v$, so that

$$\frac{\partial f_1}{\partial u} = -\cos v, \quad \frac{\partial f_1}{\partial v} = u \sin v, \quad \frac{\partial f_2}{\partial u} = -\sin v, \quad \frac{\partial f_2}{\partial v} = -u \cos v.$$

In turn, at $(x_0, y_0, u_0, v_0) = (0, 2, 2, \frac{\pi}{2})$, we get

$$J = \begin{pmatrix} \frac{\partial f_1}{\partial u} & \frac{\partial f_1}{\partial v} \\ \frac{\partial f_2}{\partial u} & \frac{\partial f_2}{\partial v} \end{pmatrix} = \begin{pmatrix} -\cos v & u \sin v \\ -\sin v & -u \cos v \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ -1 & 0 \end{pmatrix}.$$

Example 13.6 # 4 (b), continued

Furthermore, a careful calculation gives $\det J = 0 \cdot 0 - (-1) \cdot 2 = 2 \neq 0$. By the implicit function theorem, we conclude that, there exists implicit functions $u(x, y)$, $v(x, y)$, in a neighborhood of $(x_0, y_0) = (0, 2)$.